



江苏常隆农化有限公司

JIANGSU CHANGLONG AGROCHEMICAL CO.,LTD.

江苏省泰兴经济开发区团结河路8号 邮编:225400 网址:www.changlongnh.com

No.8 Tuanjiehe Road, Economic Development District of Taixing, Jiangsu, China-225400

Tel: 86-523-89710025; Fax: 86-523-87510531; E-mail:jscl@jschanglong.com

MATERIAL SAFETY DATA SHEET

1. Product And Company Identification:

Product: Carbaryl 85% WP

Company Identification: Jiangsu Changlong Agrochemical Co., Ltd.

No. 8 Tuanjiehe Road, Economic Development District of Taixing,
Jiangsu, China 225400

UN No.: 2757

CAS No.: 63-25-2

Chemical Family: Carbamate 850 g/kg

Chemical Formula: $C_{12}H_{11}NO_2$

Molecular weight: 201.2

Use: Insecticide

2. Composition/Information On Ingredients:

Ingredients	Conc. (%)
Carbaryl 95%Technical	89.47
Sinocal PW(emulsifier)	5.00
Acid Clay	5.53

3. Hazardous Identifications:

Main Hazard: Toxic to fish and bees. **Carbaryl** is a carbamate compound, which inhibits cholinesterase. It is toxic. Contact with skin, inhalation of dust or spray, or swallowing may be fatal.

Toxicity class: WHO II, EPA I

ADI: 0.01 mg/kg b.w

NOEL: 200 mg/kg (rats) - 2 year

ACGIH: 5 mg/m³ **Carbaryl**

10 mg/m³ Nuisance dust

STEL: 15 mins 10 mg/m³

TWA: 8 hours 5 mg/m³

Fire and explosion hazard: Slight fire hazard when exposed to heat or flame. Dust-air mixtures may ignite or explode.

Biological Hazard: Likely routes of exposure: May be absorbed from the gastrointestinal tract, through the intact skin, and through inhalation of fine spray mist or dust.

Eye contact: Tests indicate the product is minimally toxic; however caution should be practiced when handling the product. The product was found to be non-irritating.

Skin contact: Tests indicate the product is minimally toxic; however caution should be practiced when handling the product. The product was found to be non-irritating.

Ingestion: Toxic by ingestion. See point 4 for symptoms.

Inhalation: Toxic by inhalation. See point 4 for symptoms.



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4. First Aid Measures:

Proper care should be taken during occupational use to avoid any inhalation of dust and spray particles, and to prevent accidental contamination of food products and water. The product is a Cholinesterase inhibitor.

Inhalation:

Acute exposure: When inhaled, the first effects of cholinesterase inhibition are usually respiratory and may include nasal hyperaemia and watery discharge, chest discomfort, dyspnea, and wheezing due to increased bronchial secretions and bronchoconstriction. Other systemic effects may begin within a few minutes or several hours of exposure.

Symptoms may include nausea, vomiting, diarrhoea, abdominal cramps, headache, vertigo, ocular pain, ciliary muscle spasm, blurring or dimness of vision, miosis, or in some cases mydriasis, lacrimation, salivation, sweating, and confusion. Other reported central nervous system or neuromuscular effects include ataxia, slurred speech, weakness, fatigue, twitching, fasciculation, tremor, and eventually paralysis of the extremities and possibly of the respiratory muscles. In severe cases, there may also be involuntary defecation and urination, bradycardia, hypotension, pulmonary oedema, convulsions, coma, and death from respiratory failure or cardiac arrest. **Carbaryl** does not accumulate in mammalian tissue and the cholinesterase inhibition reverses rather rapidly. In nonfatal cases, the illness generally lasts less than 24 hours.

Chronic exposure:

Prolonged or repeated exposure may cause effects as described in acute exposure.

First aid:

Remove from exposure area to fresh air immediately. If breathing has stopped, give mechanical artificial respiration (not direct mouth-to-mouth). Maintain airway and blood pressure and administer oxygen if available.

Keep affected person warm and at rest. Treat symptomatically and supportively. Qualified personnel should perform administration of oxygen. Get medical attention immediately.

Skin contact:

Acute exposure: Some compounds may cause irritation. Localized sweating and fasciculations may occur at the site of contact. If sufficient amounts are absorbed through the skin, other effects of cholinesterase inhibition may occur as described in acute inhalation. Symptoms may be delayed for 2-3 hours, usually no more than 8 hours.

Chronic exposure:

Repeated or prolonged exposure may cause effects as described in acute exposure.

First aid:

Remove contaminated clothing immediately. Wash contaminated areas with soap and water followed by alcohol. Emergency personnel should wear gloves and avoid contamination. Treat respiratory difficulty with mechanical artificial respiration. Get medical attention immediately.



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Eye contact:

Acute exposure:

Direct contact may cause pain, hyperaemia, and lacrimation, twitching of the eyelids, miosis, and ciliary muscle spasm with loss of accommodation, blurred or dimmed vision and browache. Sometimes mydriasis may occur instead of miosis. With sufficient exposure, other symptoms of cholinesterase inhibition may occur as described in acute inhalation.

Chronic exposure:

Prolonged exposure may cause effects as described in acute exposure. Some compounds have caused toxic effects on the crystalline lens, conjunctival thickening and obstruction of nasolacrimal canals when used as miotic eye drops.

First aid:

Irrigate eyes with water or saline solution. If symptoms of poisoning occur, treat respiratory difficulty with mechanical artificial respiration and oxygen. Observe patient for at least 24-36 hours. Get medical attention immediately. Qualified medical personnel should administer oxygen.

Ingestion:

Acute exposure:

When ingested, the first effects may be nausea, vomiting, anorexia, abdominal cramps, and diarrhoea. With absorption from the gastrointestinal tract, the other effects of cholinesterase inhibition as described in acute inhalation may occur. Symptoms may begin within minutes or be delayed several hours.

Chronic exposure:

Repeated ingestion may cause effects as described in acute exposure.

First aid:

If person is alert and respiration is not depressed, give syrup of Ipecac followed by water (if vomiting occurs, keep head below hips to prevent aspiration). If consciousness level declines or vomiting has not occurred in 15 minutes empty stomach by gastric lavage with the aid of cuffed endotracheal tube using isotonic saline or 5 % sodium bicarbonate follow with activated charcoal.

Establish and maintain airway. Treat respiratory difficulty with artificial respiration and oxygen.

Do not give morphine, aminophylline, phenothiazines, reserpine, furosemide, or ethacrynic acid. Drugs like 2 PAM are not effective in poisoning with Carbaryl AND SHOULD NOT BE USED.

Treat symptomatically and supportively. Qualified medical personnel must perform administration of oxygen and gastric lavage. Get medical attention immediately.

Advice to physician:

Antidote:



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The following antidote has been recommended. However, the decision as to whether the severity of poisoning requires administration of any antidote and actual dose required should be made by qualified medical personnel.

For cholinesterase inhibitors: Establish clear airway and tissue oxygenation by aspiration of secretions, and if necessary, by assisted pulmonary ventilation with oxygen.

Improve tissue oxygenation as much as possible before administering atropine to minimize the risk of ventricular fibrillation. Administer atropine sulphate intravenously, or intramuscularly if iv injection is not possible. In moderately severe poisoning administer atropine sulphate, 0.4-2.0 mg repeated every 15 minutes, until atropinization is achieved (tachycardia, flushing, dry mouth, mydriasis). Maintain atropinization by repeated doses for 2-12 hours, or longer, depending on the severity of poisoning. The appearance of rales in the lung bases, miosis, salivation, nausea, bradycardia, are all indications of inadequate atropinization. Severely poisoned individuals may exhibit remarkable tolerance to atropine. Two or more times the dosages suggested above may be needed. Persons not poisoned or only slightly poisoned, however, may develop signs of atropine toxicity from such large dosages: fever, muscle fibrillations, and delirium are main signs of atropine toxicity. If these signs appear while the patient is fully atropinized, atropine administration should be discontinued, at least temporarily. Observe treated patients closely at least 24 hours to ensure that symptoms (possibly pulmonary oedema) do not recur as atropinization wears off. In very severe poisonings, metabolic disposition of toxicant may require several hours or days during which atropinization must be maintained.

Markedly lower levels of urinary metabolites indicate that atropine dosage can be tapered off. As dosage is reduced, check the lung bases frequently for rales. If rales are heard or other symptoms return, re-establish atropinization promptly.

5. Fire Fighting Measures:

Fire and explosion hazard: Slight fire hazard when exposed to heat or flame. Dust-air mixtures may ignite or explode.

Extinguishing agents: Extinguish small fires with carbon dioxide, dry powder, Halon, water spray, or alcohol resistant foam. Water spray can be used for cooling of unaffected stock, but avoid water coming in contact with the product. Contain water used for firefighting for later disposal

Firefighting: Move containers from fire area if possible. Fight fire from maximum distance. Stay away from storage tank ends. Contain fire control water for later disposal. Do not scatter material, extinguish only if flow can be stopped.

Use flooding amounts of water as a fog as solid streams may be ineffective. Cool containers with flooding amounts of water as far a distance as possible. Use water spray to absorb toxic vapors. Avoid breathing toxic vapors.

Keep upwind. Consider evacuation of downwind area if material is leaking.

Special Hazards: Fire may produce irritating or poisonous vapors (sulphoxides), mists or other products of combustion.

Personal protective equipment: **Carbaryl dust** may be transported in the smoke from a fire. Fire fighters and others that may be exposed should wear full protective clothing and self-contained breathing apparatus.



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6. Accidental Release Measures:

Personal precautions: Avoid contact with skin and eyes.

Do not breathe in dust or fumes. For personal protection see Section 8.

Environmental precautions: Do not allow entering drains or watercourses. When the product contaminates public waters, inform appropriate authorities immediately in accordance with local regulations.

Occupational spill: Do not touch spilled material. Stop leak if you can do so without risk. Use water spray to reduce vapors (contain any water used). For **small spills**, sweep up with sand or other suitable absorbent material, such as sawdust, and place into containers for later disposal. Move containers from spill area. For **larger spills**, contain material far ahead of spill for later disposal.

Keep spectators away. Isolate hazard area and deny entry. Ventilate closed spaces before entering.

7. Handling and Storage:

Handling: Toxic by inhalation or if swallowed. Avoid contact with eyes, prolonged contact with skin, and inhalation of dust and vapor. Use with adequate ventilation. Wash hands before eating, drinking, chewing gum, smoking, or using the toilet. Remove clothing immediately if this product gets inside. Then wash skin thoroughly using a non-abrasive soap and put on clean clothing. Do not apply directly to areas where surface water is present, or to intertidal areas below the mean high water mark. Water used to clean equipment must be disposed of correctly to avoid contamination.

Storage: The product must be kept under lock and key. Keep out of reach of unauthorized persons, children and animals. Store in its original labeled container in shaded, well-ventilated area, away from heat, sparks and other sources of ignition. Not to be stored next to foodstuffs and water supplies. Local regulations should be complied with.

8. Exposure Control/Personal Protection:

Occupational exposure limits: No occupational limits established by OSHA, ACGIH or NIOSH

Engineering control measures: It is essential to provide adequate ventilation. The measures appropriate for a particular worksite depend on how this material is used and on the extent of exposure. Ensure that control systems are properly designed and maintained. Comply with occupational safety, environmental, fire, and other applicable regulations.

Personal Protective Equipment:

If engineering controls and work practices are not effective in controlling exposure to this material, then wear suitable personal protective equipment including approved respiratory protection.

Respirator: An approved respirator suitable for protection from dusts and mists of pesticides is adequate. Limitations of respirator use specified by the approving agency and the manufacturer must be observed.

Clothing: Employee must wear appropriate protective (impervious) clothing and equipment to prevent repeated or prolonged skin contact with the substance.



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Gloves: Employee must wear appropriate synthetic protective gloves to prevent contact with this substance.

Eye protection: The use of full-face protection is recommended.

Emergency eyewash: Where there is any possibility that an employee's eyes may be exposed to this substance; the employer should provide an eye wash fountain or appropriate alternative within the immediate work area for emergency use.

9. Physical and Chemical Properties:

Appearance:	White powder.
Odor:	Odorless.
Flammability:	Not flammable.
Explosive properties:	Like most organic powders, this product could form explosive mixture in air, under severe dust conditions.
Flash point:	193°C (data for technical material).
Specific gravity:	0.55 g/ml
Oxidizing properties:	Not oxidative.
pH:	6.5
Stability:	Stable in neutral and acidic media, but hydrolyzed in alkaline media to 1-naphthol. Rapidly converted by oxidizing agents. Stable in light and heat.
Persistent foaming:	Not available.
Solubility in water:	The product is a wettable powder.

10. Stability and Reactivity

Stability: Stable up to 2 years under normal storage conditions. Stable in neutral and acidic media, but hydrolyzed by concentrated alkalis to form 1-naphthol.

Half-life is 12 days (pH 7) and 3.2 days (pH 9). The rate of decomposition increases at higher temperatures. **Carbaryl** is stable to light and heat.

Incompatibility: Compatible with most insecticides, fungicides and acaricides. Alkaline substances such as lime and Bordeaux mixture may reduce the activity of the product. The product should therefore not be used if the soil or water has high pH values.

Hazardous decomposition: Toxic oxides of nitrogen are released when the product decomposes on heating.

11. Toxicological Information

All data is for technical material.

Acute oral: LD₅₀ (rats) was more than 300 to less than 2000 mg/kg bodyweight.

Acute dermal: LD₅₀ (rats) was estimated to be greater than 2000 mg/kg bodyweight.

Inhalation toxicity: LC₅₀ (rats) was estimated to be greater than the maximum attainable concentration of 3.51 mg/L.

Skin irritation: Non Irritant to skin of rabbit.

Eye irritation: Minimal Irritant to eye in rabbit.

Skin or dermal sensitization: Non Sensitizer to guinea pig.



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Carbaryl 85% WP

Acute oral toxicity: LD₅₀ (rats) was more than 300 to less than 2000 mg/kg bodyweight.

Acute dermal: LD₅₀ (rats) was estimated to be greater than 2000 mg/kg bodyweight.

Inhalation toxicity: LC₅₀ (rats) was estimated to be greater than the maximum attainable concentration of 3.99 mg/L.

Skin irritation: Non Irritant to skin of rabbit.

Eye irritation: Minimal Irritant to eye in rabbit.

Skin or dermal sensitization: Non Sensitizer to guinea pig.

Carcinogenicity: Evidence indicates that **Carbaryl** is unlikely to be carcinogenic in humans.

Teratogenicity: Evidence for teratogenic effects due to chronic exposure is minimal in test animals.

Mutagenicity: **Carbaryl** has been shown to affect cell division and chromosomes in rates. However, evidence suggests that **Carbaryl** is unlikely to be mutagenic to humans.

12. Ecological Information:

Degradability: In soil, the active ingredient is metabolized to form 1-naphthol. The half-life of the product is 7-14 days in sandy loam soils and 14-28 days in clay loam.

Soils with high organic matter content retain residues for longer periods than do mineral soils.

Mobility: The product is adsorbed on soil and is unlikely to leach into water sources.

Accumulation: The product adsorbs to soil but shows little or no tendency to bio-accumulate. **Carbaryl** has very limited persistence in the environment.

Ecotoxicology:

Birds: Minimally toxic to birds.

Acute oral LD₅₀: > 2179 mg/kg (young mallard ducks).

> 2230 mg/kg (Japanese quail).

> 2000 mg/kg (young pheasants).

1000 - 3000 mg/kg (pigeons).

Fish: Toxic to fish.

LC₅₀ (96 hr): 1.3 mg/l (rainbow trout).

10 mg/l (bluegill sunfish).

2.2 mg/l (sheepshead minnow).

Bees: Toxic to bees.

LD₅₀ (topical): 1 mg/bee.

Daphnia: Very toxic to Daphnia.

Daphnia magna: EC₅₀ (48 hours): 0,006 mg /l.



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Earthworms:

Toxic for earthworms.

Beneficial insects:

Toxic to beneficial insects.

Soil micro-organisms:

Various soil fungi are able to metabolize **Carbaryl** and in soils previously treated with carbamates and cloethocarb, 80% of **Carbaryl** was completely mineralized to carbon dioxide during a four-week incubation period.

13. Disposal Considerations

Pesticide disposal: Contaminated absorbents, surplus product, etc., should be burned at 1000°C in a high temperature incinerator with effluent gas scrubbing. Where no incinerator is available, hydrolysis under alkaline conditions (pH 12 or above) is a suitable method to dispose of small quantities of the product. Before disposal of the resultant waste, the material must be analyzed to ensure that the active ingredient has been degraded to a safe level. Never pour untreated waste or surplus products into public sewers or where there is any danger of run-off or seepage into water systems. Comply with local legislation applying to waste disposal.

Package product wastes: If container is broken, handle with rubber gloves. Emptied containers retain vapor and product residues. Observe all labeled safeguards until container is destroyed. Combustible containers should be disposed of in pesticide incinerators. Non-combustible containers must be punctured and transported to a scrap metal facility for recycling or disposal.

14. Transport Information

UN Number 2757

ADR/IRD

Substance name: Carbamate pesticide, solid, toxic (**Carbaryl** 850 g/kg)

Label: 6.1

IMDG/IMO

Packaging group: III

Label of class: 6.1 **Marine pollutant**

Shipping name: Carbamate pesticide, solid, toxic (**Carbaryl** 850 g/kg).

AIR/IATA

Shipping name: Carbamate pesticide, solid, toxic (**Carbaryl** 850 g/kg).

Class: 6.1

Hazard Label: Toxic

Packaging Group: III

Passenger Aircraft: Y 619 (10kg), 619 max 100 kg)

Cargo Aircraft: 619 (max 200 kg).

15. Regulatory Information

Symbol: Xn, N



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Indication of danger: Harmful, Environmentally Hazardous Substance

Risk phrases:

R 22 Harmful if swallowed

Safety phrases:

S 2 Keep out of reach of children.

S 22 Avoid breathe dust.

S 24 Avoid skin contact.

S 36/37 Wear suitable protective clothing and gloves.

S 46 If swallowed, seek medical advice immediately and show label or MSDS.

S 61 Avoid release to the environment. Refer to special instructions on the label and MSDS.

16. Other Information:

N/A

ORIGINAL CONTROLLED